Nessma Mohdy

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Experience

Platform Software Intern @ Figure AI *Austin, Texas* | September – December 2026

- Owned **end-to-end teleoperation integration** across Linux applications and system services, validating **5G, Wi-Fi, and wired** communication, **RTOS scheduling**, and SDK integration using **C++** and **Python**. Analyzed kernel and system logs with `journalctl`, investigated network configurations with `ip` and `nmcli`, and used `tcpdump` captures for script-driven analysis and reproduction of communication events to root-cause and resolve integration failures, achieving **100% nightly test passes**.
- Developed an **end-to-end latency measurement and analysis pipeline** spanning robot-to-teleoperation communication and peripheral devices, including hand-tracking hardware, IMUs, and **Meta Quest headsets**. Correlated frame-level image data with **MCAP recordings** to characterize latency and identify performance bottlenecks across the system.
- Built a **distributed Wi-Fi testing system** using configurable routers to dynamically enable and disable connectivity, enabling controlled validation of teleoperation behavior under varying network conditions.

Vehicle Software Intern @ Tesla *Palo Alto, California* | January – August 2026

- Developed a **layered automotive diagnostic communication stack** following the OSI model, implementing **SAE J1979/ISO 15031-5 OBD-II** support over **ISO 15765-2 (ISO-TP)** and **CAN (DoCAN)** through a proxy architecture scalable across vehicle platforms. Integrated the stack on an RTOS to meet the **50 ms P2 tester timing budget** and validated reliability through **fuzz testing** and **long-duration soak tests**.
  - Enabled **Diagnostic Trouble Code (DTC) support** on the Cybertruck main vehicle controller by integrating the Diagnostic Event Manager (DEM) library with **Unified Diagnostic Services (UDS)** hooks.
  - Investigated C compilation and linking behavior to support identically named functions across vehicle-specific builds, analyzing **symbol resolution** and linker behavior to prevent naming conflicts between platform variants.
- Closed a **DFMEA-flagged Steer-by-Wire safety gap** by gating steering authority on controller qualification across **5 controllers and 2 monitors**, persisting fault state in **NVRAM** across power cycles, and adding runtime rear-rack health checks (rolling counters, interrupt-driven timeouts) that propagate a rear-rack failure to every controller.
  - Rejected stale rear-rack SENT frames with an incrementing-counter check and tuned SPC58 eMIOS input filtering, cutting noise-induced signal errors by **30% under simulated EMI**.
- Redesigned Cybercab's steer-by-wire-derived electronic parking brake status logic into a **table-driven cross-check** that resolves brake state from independent actuator and controller signals, covering every input combination deterministically and removing reliance on a mechanical steering-column fallback and parking-pawl position signal.
- Built a **Rust model of a 3-phase gate driver** (register map, SPI framing, fault injection) with a Python test interface for **SIL**.
- Managed build and dependency isolation with **Buck2**, and extended a **Haskell code generator** to eliminate a class of **configuration-drift bugs** between JSON configs and C MIA handlers across multiple ECU configurations.

Firmware Development Intern @ AMD *Markham, Ontario* | May – December 2025

- Developed **platform-agnostic x86 firmware in C** bridging core boot and **UEFI**, integrating GPIO and NBIO subsystems for the **Asus ROG Xbox platform**, enabling cross-platform deployment across consoles and servers with minimal code duplication.
- Received the **Spotlight Award** for building a cross-platform Python/Unix application to automate pre-silicon simulation updates, optimizing power and storage consumption by approximately **20GB per employee per day**.
- Engineered an **automated triage system** using JIRA and Synopsys APIs in Python, reducing a 2-hour manual debugging process to **5 minutes** and clearing **3,000+ issues within 2 months**.

Machine Vision Engineer Intern @ Tesla *Austin, Texas* | January – May 2025

- Initiated and deployed an **end-to-end machine vision system** for cathode production, owning camera selection, setup, and installation, and implementing a **ResNet-based model** for real-time defect and crack detection with **94%+ accuracy**, helping prevent **4 days of downtime and $5M USD** in collateral losses.
  - Benchmarked ResNet, Vision Transformer, YOLO, and a custom ResNet using **PyTorch** and **CUDA** to optimize accuracy-latency trade-offs for production deployment.
- Co-developed a **Flask, InfluxDB, and Caspian monitoring platform** to visualize thousands of cells processed weekly, search historical production data, and display heatmaps. Integrated Halcon image analysis via FTP and implemented **query caching and incremental retrieval** to accelerate data loading.

Projects

Wikipedia Race Game | *JavaScript, Flask/PostgreSQL, WebSocket, ONNX Runtime*

- Developed a **full-stack multiplayer Wikipedia racing game** with a Chrome extension for timed gameplay and leaderboards. Building Flask/PostgreSQL backend with WebSocket communication for real-time co-op, and designing an **ML difficulty engine** using ONNX Runtime, and **Dijkstra's algorithm** based on category divergence.

C Compiler | *C, x86-64 Assembly*

- Built a **C compiler entirely from scratch in C**. Including hand-written lexer, **recursive-descent parser**, and a **three-address IR** supporting a substantial C subset of control flow, functions, integer/unsigned/floating-point types, pointers, arrays, structs, and dynamic memory with a backend emitting **x86-64 assembly**.

Technical Skills

Languages: Python, C/C++, Rust, TypeScript/JavaScript, Java, SQL, MATLAB, Assembly
AI / ML: PyTorch, ONNX Runtime, CUDA, ResNet, Vision Transformer, YOLO, MCP, Reinforcement Learning, ROS2